GROWMATE EDU: Startup Execution Plan

Learn. Grow. Preserve. Nourish.

Prepared by: Hareem Fatima and team | Status: living document, review monthly

Note on numbers: every figure marked (A) is an assumption to be verified. Figures marked (V) must be filled from a cited source before final submission.

Executive Summary

Problem: Schools and households in Pakistan lack practical, locally relevant, affordable education on growing food, seed preservation and nutrition. This knowledge sits in separate apps, books and NGO programmes.

Solution: GROWMATE EDU is an AI-powered education platform that takes learners from seed selection to growing, harvest, preservation and nutrition science, adapted to their city, climate and level.

Target customers: Private schools (primary), urban households (secondary), NGOs and universities later.

90-day goal: Working MVP piloted with 3 schools and 50 households, 10 expert-reviewed Karachi crop guides, 5 signed letters of intent.

12-month goal: 15 paying schools, 600 paying households, PKR 3M revenue (A).

Needs: Seed funding of PKR 25M, about USD 90k at 280 PKR/USD (A). People: CTO, agronomist advisor, school partnerships lead. Partners: 3 pilot schools, one agriculture university, one NGO.

Problem and Market

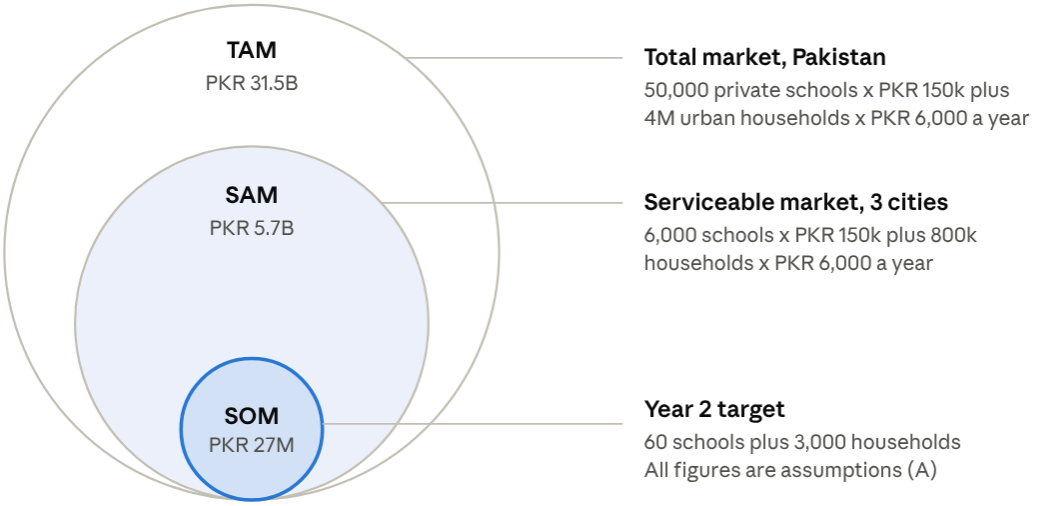
Who has the problem: Urban parents who want healthier eating and activity for children, teachers lacking practical science resources, and beginner growers in heat-stressed cities such as Karachi.

Pain level: Moderate. A nice to have for households, a curriculum need for schools. This is the main risk to manage.

TAM, SAM, SOM (A)

Layer	Definition	Value
TAM	Pakistan: 50,000 private schools x PKR 150k plus 4M urban households x PKR 6,000 per year	about PKR 31.5B
SAM	Karachi, Lahore, Islamabad: 6,000 schools x PKR 150k plus 800k households x PKR 6,000	about PKR 5.7B
SOM	Year 2 target: 60 schools plus 3,000 households	about PKR 27M

The reachable market narrows from PKR 31.5B to a PKR 27M target



TAM, SAM, SOM · 3 layers

Key statistics to add (V)

Urban population share, household food insecurity rate, number of private schools, smartphone and internet penetration, child malnutrition prevalence. Sources: Pakistan Bureau of Statistics, UNICEF, National Nutrition Survey, school census data.

Why seed literacy matters now

Seeds are getting harder to save, and that is the long term case for teaching seed preservation.

- Hybrid (F1) seeds do not breed true, so saved seed gives unreliable plants and growers must buy new seed every season (Organic Valley).

- Patented and genetically modified seeds are often sold under licences that bar saving and replanting (Non-GMO Project).
- Seedless fruit and vegetables, such as seedless watermelon, grapes and citrus, are a growing consumer preference. Home growers cannot save seed from them to grow the next crop.
- Advocacy groups report that agricultural biodiversity has fallen sharply as hybrid seed spread, and that seed saving skills are being lost (Sustainable Food Trust).

Our position: GROWMATE EDU teaches households and schools to save seed from open pollinated and heirloom varieties, so they keep choice, local adaptation and lower seed costs.

What we do not claim: seeds will not disappear, and hybrid or improved seed has real benefits for farmers. The claim is narrower: access to seed that can be legally and reliably saved is shrinking, and few learners are taught how to do it. Seed rules differ by country and variety, so we teach education only and sell no seed until legal review (see Legal, Regulatory and Risk).

Solution and Product Scope

MVP features (Phase 1)

1. AI Grow Coach: step by step guidance for beginners.
2. Location and weather recommendations: manual city selection, forecast based advice.
3. Karachi planting calendar.
4. Digital Plant Passport: record of each plant's growth and harvest.
5. Nutrient Finder and Food Combination Checker (basic).
6. School and teacher dashboard.

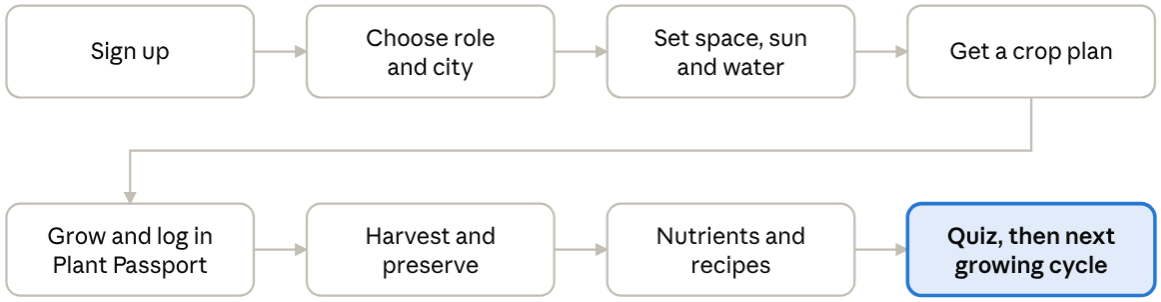
Not building yet

Plant Doctor image diagnosis, Seed Guardian, smart sensors, community network, marketplace, seed sales, educational kits.

User journey

Sign up, choose role and city, set space, sunlight and water, receive crop plan, grow and log in Plant Passport, harvest, explore nutrients and recipes, take quiz, start next cycle.

Every learner follows one loop from sign up to the next growing cycle



user journey · 8 steps

Feature roadmap

Phase	Timing	Scope
1	Months 1 to 4	MVP six features
2	Months 5 to 10	Plant Doctor, Seed Guardian, School Garden Lab, Food Science Academy
3	Months 11 to 24	More cities and countries, Urdu voice, offline mode, sensors, expert network

Competitor and Positioning Analysis

Competitor	Focus	Gap versus GROWMATE EDU
Plantix	Crop disease for farmers	Not household, school or nutrition focused
Ayki	AI garden planning	No school curriculum or nutrition science
From Seed to Spoon	Vegetable growing	No assessment or nutrient interaction learning
Cropanion	Climate aware gardening	Narrow scope, no school layer
Crops in Pots	School gardening, offline	Not scalable digitally
Organica Pakistan	Agroecology training	Limited AI personalisation

Competitor	Focus	Gap versus GROWMATE EDU
Nutrition tracking apps	Calories and macros	No growing link, shallow nutrient interaction content

Advantage: founder expertise in clinical nutrition and functional foods, plus one connected grow to plate curriculum.

2x2 positioning matrix

	Weak growing education	Strong growing education
Strong nutrition education	Nutrition apps	**GROWMATE EDU**
Weak nutrition education	Generic AI chat	Plantix, Ayki, Cropanion

Business Model and Pricing

Stream	Price	Notes
School licence	PKR 150,000 per year (about PKR 300 per student, 500 students)	Priority stream
Household premium	PKR 499 per month	Free tier for basics
Workshops	PKR 40,000 per session	Low volume
Partnerships	Custom	Year 2 onwards

Unit economics (A)

Metric	Household	School
Revenue	PKR 499 per month	PKR 150,000 per year
Gross margin	75%	70%
CAC	PKR 1,500	PKR 25,000
Churn	8% per month	20% per year
LTV	about PKR 4,680	about PKR 525,000

Metric	Household	School
LTV to CAC	about 3.1	about 21

School figures are optimistic until a school actually pays. Validate in the pilot.

Go to Market Plan

Channels: direct school outreach (principals, science heads), parent WhatsApp groups, Instagram and TikTok gardening and nutrition content, university nutrition and agriculture societies, NGO partnerships.

Early customers: 3 pilot schools, free for one term in exchange for feedback and a testimonial, then paid conversion.

Message: From seed to science. Your child grows food and learns nutrition.

Funnel, 12 month targets (A)

Stage	Target	Conversion
Awareness	20,000 reach	5%
Signup	1,000 free users	40%
Activation	400 complete first plant plan	15%
Payment	60 household payers plus schools via direct sales	



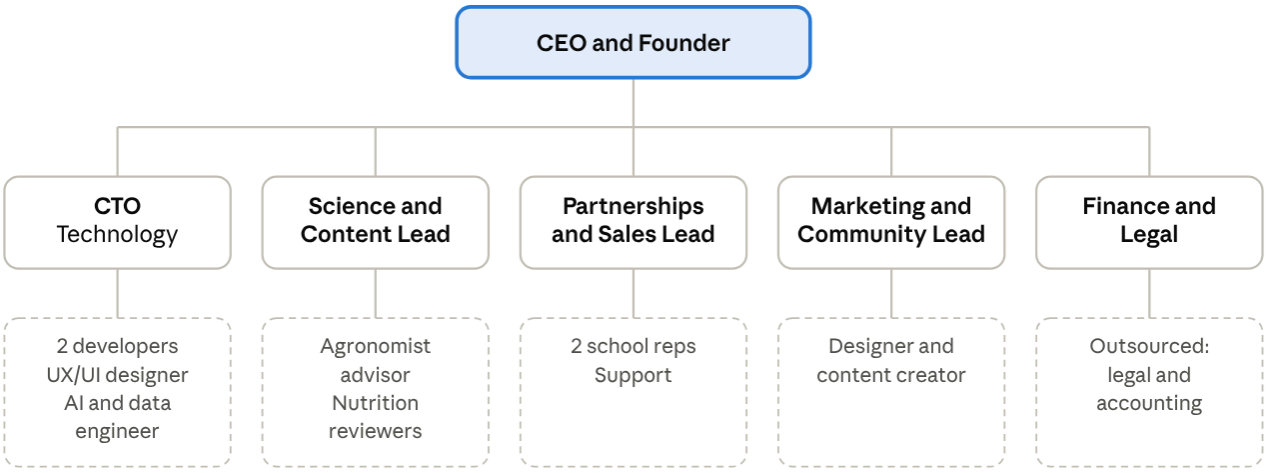
conversion funnel · 4 stages

Operations and Team

Organisation chart

Level	Role	Reports to
1	CEO and Founder	Board and advisors
2	CTO (Tech)	CEO
2	Science and Content Lead	CEO
2	Partnerships and Sales Lead	CEO
2	Marketing and Community Lead	CEO
2	Finance and Legal (outsourced)	CEO
3	2 Developers, UX/UI designer, AI and data engineer	CTO
3	Agronomist advisor, nutrition reviewers	Science Lead
3	2 school representatives, support	Sales Lead
3	Designer and content creator	Marketing Lead

Five leads report to the founder; finance and legal are outsourced



Solid box = lead role. Dashed box = team and advisors under that lead.

organisation chart · 2 levels

Tools: Lovable or Bolt for prototype, Supabase, weather API, AI API with retrieval from verified content, Canva, Notion, WhatsApp Business.

Outsourced: legal, accounting, early development sprints, Urdu translation.

Key hires: CTO (month 1), agronomist advisor part time (month 1), school sales lead (month 3).

Legal, Regulatory and Risk

Risk	Impact	Mitigation
Seed law and plant variety rules	High	Education only, no seed sales until legal review
Wrong nutrition or plant advice from AI	High	Expert review, source citations, confidence levels, disclaimers
Child data and photo privacy	High	Parental and school consent, minimal data, secure storage
Weak school willingness to pay	High	Pilot first, test price early
AI API cost increase	Medium	Caching, tiered usage, cheaper models for simple queries
Content accuracy at scale	Medium	CMS with expert approval workflow
Dependence on founder	Medium	Document processes, hire early
Low household retention	Medium	Weekly tasks, reminders, school to home assignments

Legal to do: register company with SECP, check data protection obligations, draft Terms and Privacy Policy, confirm medical advice boundaries. Have a lawyer review.

Execution Timeline

30, 60, 90 days

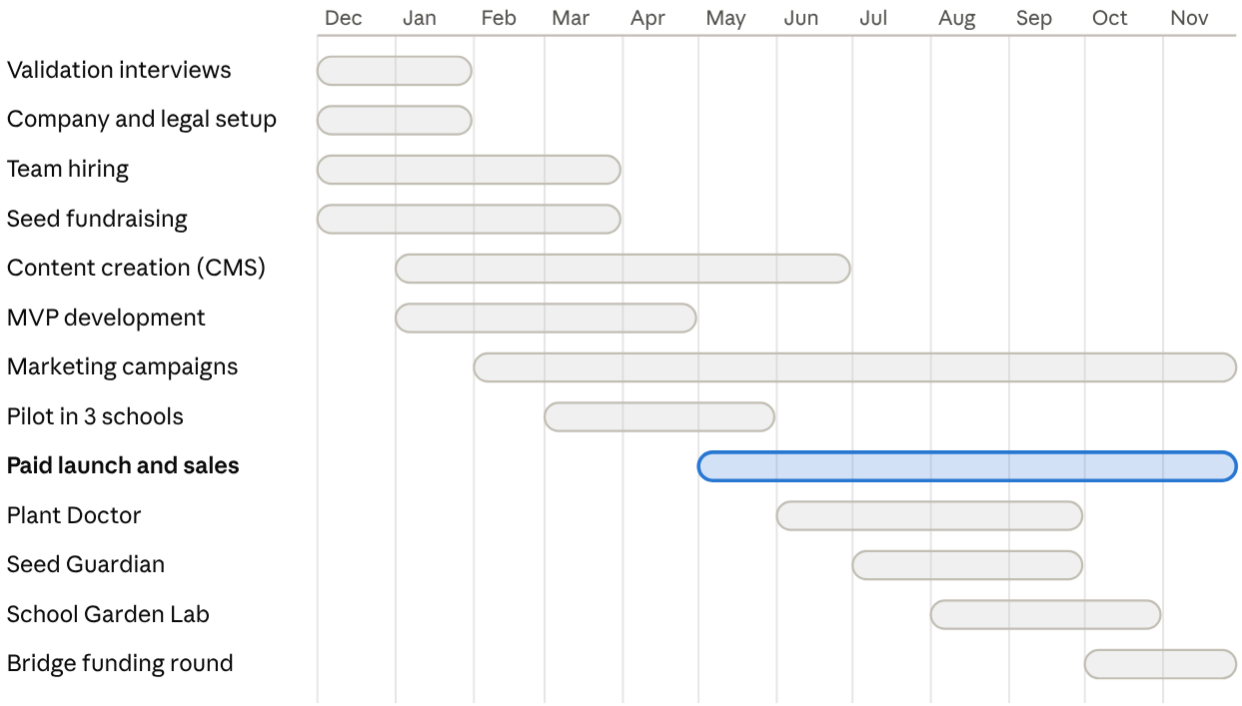
Window	Milestone	Owner
Days 1 to 30	Interviews (10 teachers, 20 households, 5 experts), company registration, hire CTO	CEO
Days 31 to 60	MVP build, 10 Karachi crop guides expert reviewed	CTO, Content Lead
Days 61 to 90	Pilot in 3 schools and 50 households, 5 letters of intent	Sales Lead

12 month Gantt chart (■ = active)

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Validation interviews	■	■										
Company and legal setup	■	■										
Team hiring	■	■	■	■								
Content creation (CMS)		■	■	■	■	■	■					
MVP development		■	■	■	■							
Pilot (3 schools)				■	■	■						
Plant Doctor (Phase 2)							■	■	■	■		
Seed Guardian								■	■	■		
School Garden Lab									■	■	■	
Paid launch and sales						■	■	■	■	■	■	■
Marketing campaigns			■	■	■	■	■	■	■	■	■	■
Seed fundraising	■	■	■	■							■	■
Monthly KPI review	■	■	■	■	■	■	■	■	■	■	■	■

Paid launch starts in month 6, right after the school pilot

12 month plan, assumed start December 2026



12 month Gantt chart · 13 workstreams

Financial Plan (PKR millions)

Year 1 budget

Item	PKR M
Team salaries	11.0
Product development and AI/API	6.0
Marketing and school pilots	3.5
Content and expert review	1.5
Operations and tools	1.2
Legal and compliance	0.8
Contingency	1.0
Total	**25.0**

24 month forecast

Quarter	Revenue	Costs	Net
Q1	0.0	5.0	-5.0
Q2	0.4	6.0	-5.6
Q3	1.0	6.5	-5.5
Q4	1.6	7.5	-5.9
Q5	3.0	8.0	-5.0
Q6	4.5	8.5	-4.0
Q7	6.5	9.0	-2.5
Q8	8.5	9.5	-1.0

Break even: around month 28 to 30. **Runway:** PKR 25M covers about 12 months, so a bridge round of PKR 15M to 20M is needed in months 10 to 12.

Metrics and KPIs

KPI	12 month target
Signups	1,000
Activation (first plan completed)	40%
Paying households	600
Paying schools	15
Monthly household churn	Under 8%
Household CAC	Under PKR 1,500
30 day retention	35%
Revenue	PKR 3M
Expert reviewed content items	200

Next Actions

- [ ] Finalise concept and name check (CEO, this week)

- [ ] Complete interviews: 10 teachers, 20 households, 5 experts (CEO and team, days 1 to 30)
- [ ] Register company and open bank account (Finance and Legal, day 30)
- [ ] Recruit CTO and agronomist advisor (CEO, day 30)
- [ ] Build clickable MVP prototype (CTO, day 60)
- [ ] Expert review 10 Karachi crop guides (Content Lead, day 60)
- [ ] Sign 3 pilot schools (Sales Lead, day 75)
- [ ] Launch pilot (all, day 90)
- [ ] Prepare investor deck and begin seed outreach (CEO, day 90)

Appendix. Final Positioning

Business category: AI, AgriTech, EdTech, FoodTech, Nutrition Education.

One line pitch: GROWMATE EDU transforms how people learn about food by combining AI powered agricultural guidance, climate intelligence, seed preservation education, food science and nutrient interaction learning into one connected platform, from seed to soil and from harvest to plate.